Iteration 46

Frontend Design System and Prototype

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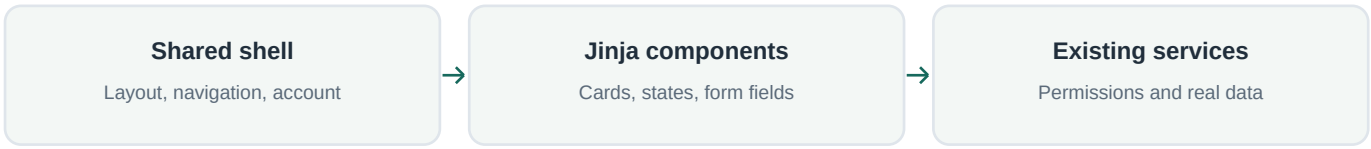
Review boundary: working visual prototype, not a completed application-wide redesign. Shared chrome, student home and one professor-appointment form are ready for review. No merge, deployment or wider rollout is included.

1. Objective and audit priorities

Reduce visual inconsistency and navigation overload without changing requests, permissions or scheduling decisions. The audit inspected all 44 existing templates, the legacy and calendar stylesheets, four frontend scripts and their route/service boundaries.

Priority	Finding and response
High	Crowded wrapping navigation: replace with a role-aware sidebar, compact header and accessible phone drawer.
High	Dashboard information overload: expose real counts, next meeting and service actions first; retain lists and the expandable month/week timetable.
High	Unstructured appointment entry: group participants, meeting details and feasible time selection; improve validation, selected-time and loading feedback.
Medium	Inconsistent cards, badges and forms: establish reusable components now; migrate other role pages only after visual approval.

2. Architecture and reusable components



PostgreSQL persistence and C++ JSON scheduling/allocation remain unchanged.

Component	Responsibility
base / navigation / account	Shared authenticated shell, permitted navigation, initials avatar, notification bell and CSRF-protected account actions.
tokens.css / shell.css / shell.js	Namespaced colors, spacing, shape and typography; responsive chrome, drawer focus containment and Escape restoration.
ui.html / prototype.css	Reusable icons, cards, buttons, badges, alerts and field styling; opt-in prototype page styles.
student_home / appointment_new	Existing private dashboard data and request payloads, with clearer hierarchy; no new database queries or business rules.

3. Design and preserved behavior

A restrained slate-teal sidebar, neutral workspace and white surfaces replace decorative gradients and competing pill navigation. Local typography and inline icons require no remote fonts/CDN. Core checked text/state contrast ranges from 5.55:1 to 13.86:1. Status labels do not rely on color alone.

Tabler informed layout and component hierarchy only. Installing Bootstrap over existing global CSS was rejected to avoid cascade collisions and a large migration. Existing legacy page interiors stay supported. Template composition reuses the established presentation/service separation; no new GoF pattern or frontend framework is needed.

Real participant-scoped counts, private meeting links, optional invitations, room history, calendar links and timetable navigation remain available. The appointment form retains all endpoints/fields, professor filtering, one date, manual proposals, C++ recommendations, server validation and CSRF.

4. Executed verification

Check	Actual result
Full Python / PostgreSQL	322 run: 321 passed, 0 failed, 1 expected Windows skip for POSIX executable permissions. Includes strategy parity, JSON, migrations, concurrency and recovery.
Final focused prototype tests	8 passed, 0 failed after final markup assertions.
Fresh C++ and isolated CLI	172 passed, 0 failed; 2 CLI smoke scenarios passed.
Existing authenticated routes	32 passed across student, professor and administrator; runtime file access forbidden. Staff covered in isolated tests.
Browser acceptance	Student login, unavailable date, recommendations, validation, submit, professor notification/approval and updated dashboard/timetable verified with disposable data.
Responsive / keyboard	1440, 768, 390 and 320px checked. No prototype page-level overflow; drawer focus/Tab/Escape, account, bell and skip target verified.
Integrity and syntax	All Jinja, changed JS, Python test and SVG syntax checks passed. All 48 primary table fingerprints and 4 preexisting generated CSV hashes unchanged.

5. Demonstrated scenario and next checkpoint

An isolated student selected a faculty, professor and optional invitee, chose an engine-verified Monday slot and submitted. Missing purpose showed a labelled validation error. After valid submission, the professor received the existing notification and approved. The student dashboard then showed confirmed = 1 and pending = 0; PostgreSQL stored two mandatory participant commitments.

After visual approval: refine the prototype, then migrate student list/detail pages, professor and faculty desks, and administrative tables/forms in separate tested batches. Retire legacy CSS only when its callers have migrated.

Limitations: remaining page interiors are intentionally unchanged. Browser checks used local Chromium viewport emulation, not physical devices or full screen-reader certification. No standalone CSS linter was installed. Remote CI was not run for this local checkpoint. The full frontend rollout has not started.